THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra PartI

May 15, 2021 TIME: 2:30 p.m.-4:30 p.m.(2 parts in total)

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Part I (Short Questions. Students will answer these questions on Exam Moodle directly. Questions and answers in each question will be shuffled. 3 marks for each question, 54 marks in total.)

1.

For a homogeneous system of k linear equations with $k+1$ unknowns ($k \in \mathbb{N}$), which of the following is (are) possible?

Select one or more:

- ☐ a. The solution set is empty.
- ☐ b. The solution set contains trivial the solution only.
- ☐ c. The solution set contains non-trivial solutions only.
- ☐ d. The solution set contains infinitely many solutions.

2.

If $a \neq 0$ and $a \neq \pm 1$ ($a \in \mathbb{R}$) , which of the following matrices must be invertible?

Select one or more:

- ☐ a. $\begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$
- ☐ b. $\begin{bmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{bmatrix}$
- ☐ c. $\begin{bmatrix} 0 & a & 0 \\ a & 0 & a \\ 0 & a & 0 \end{bmatrix}$
- ☐ d. $\begin{bmatrix} a & 0 & 1 \\ 0 & a & 1 \\ 1 & 0 & a \end{bmatrix}$

3.

If A is an $n \times n$ upper-triangular matrix, which of the following must be true?

Select one or more:

- ☐ a. A is diagonalizable.
- ☐ b. A has n eigenvalues (counting multiplicities).
- ☐ c. A has n linearly independent eigenvectors.
- ☐ d. $\det A$ is equal to the product of diagonal entries of A.

4.

Which of the following must be a linear combination of the vectors $(1, a, 0)^T$ and $(0, 1, b)^T$?

Select one or more:

- ☐ a. $(0, 0, 0)^T$
- ☐ b. $(1, 0, -ab)^T$
- ☐ c. $(1, a, b)^T$
- ☐ d. $(1, 0, ab)^T$

5.

If A and B are both $n \times n$ matrices with rank k, which of the following must be true?

Select one or more:

- ☐ a. rank$(A + B) \leq k$.
- ☐ b. rank$(A + B) \geq k$.
- ☐ c. rank$(A - B)$ =rank$(A + B)$.
- ☐ d. rank$(A + B)$ =rank$(A^T + B^T)$.
- ☐ e. rank(AB) =rank(BA).
- ☐ f. rank$(AB) = k$.

6.

Let A and B be square matrices of the same size. If A and B are both _____ , so is AB.

Select one or more:

- ☐ a. invertible.
- ☐ b. symmetric.
- ☐ c. upper-triangular.
- ☐ d. nonsingular.
- ☐ e. diagonalizable.

7.

Find the smallest value of k such that $[\,k \quad 2 \quad k\,]^T$ is a linear combination of $[\,1 \quad 3 \quad k\,]^T$ and $[\,1 \quad 2 \quad 3\,]^T$. (Round your answer to 2 decimal places.)

Answer:

8.

Let A be an $m \times n$ matrix. If the system $Ax = \mathbf{b}$ has a unique non-zero solution, which of the following is (are) possible?

Select one or more:

- ☐ a. $m < n$ and $\mathbf{b} = 0$
- ☐ b. $m < n$ and $\mathbf{b} \neq 0$
- ☐ c. $m > n$ and $\mathbf{b} = 0$
- ☐ d. $m > n$ and $\mathbf{b} \neq 0$

9.

Let $f(x, y) = 4 + x^3 + y^3 - 3xy$. Which of the following is (are) correct?

Select one or more:

- ☐ a. (0,0) is a saddle point.
- ☐ b. (1,1) is a relative maximum point.
- ☐ c. (0,0) is a relative minimum point.
- ☐ d. (1,1) is a saddle point.

10.

Evaluate $\iint_D 2xydA$, where D is the region lying in the first quadrant, between the circles centered at the origin with radius 2 and 5. (Round your answer to 2 decimal places.)

Answer:

11.

Find the volume of the region that lies inside $z = x^2 + y^2$ and below the plane $z = 16$. (Round your answer to 2 decimal places if it is not an integer.)

Answer:

12.

Evaluate the integral $\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{0} \cos(x^2 + y^2)dydx$. (Round your answer to 2 decimal places if it is not an integer.)

Answer:

13.

Evaluate the integral $\iint_D 6x^2 - 40ydA$, where D is the triangle with vertices $(1, 1), (3, 3)$ and $(5, 3)$. (Round your answer to 2 decimal places.)

Answer:

14.

Find the maximum value of $f(x, y) = xy$ on the ellipse $x^2 + 2y^2 = 1$. (Round your answer to 2 decimal places.)

Answer:

15.

Calculate the distance from a point $(1, 1, 1)$ to a plane $3x + 4y + 8z = 5$.

(Round your answer to 2 decimal places.)

Answer:

16.

Convert $\int_0^3 \int_0^{\sqrt{9-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{18-x^2-y^2}} x^2 + y^2 + z^2 dzdxdy$ into spherical coordinates.

Select one:

- a. $\int_0^{\pi/2} \int_0^{\pi/4} \int_0^{3\sqrt{2}} \rho^4 \sin\phi d\rho d\phi d\theta$
- b. $\int_0^{\pi} \int_0^{\pi/4} \int_0^{3\sqrt{2}} \rho^4 \sin\phi d\rho d\phi d\theta$
- c. $\int_0^{\pi} \int_0^{\pi/4} \int_0^{3\sqrt{2}} \rho^2 d\rho d\phi d\theta$
- d. $\int_0^{\pi/2} \int_0^{\pi/4} \int_0^{3\sqrt{2}} \rho^2 d\rho d\phi d\theta$
- e. $\int_0^{\pi/2} \int_0^{\pi/2} \int_0^{3\sqrt{2}} \rho^4 \sin\phi d\rho d\phi d\theta$
- f. $\int_0^{\pi} \int_0^{\pi/2} \int_0^{3\sqrt{2}} \rho^4 \sin\phi d\rho d\phi d\theta$
- g. $\int_0^{\pi} \int_0^{\pi/4} \int_0^{18} \rho^4 \sin\phi d\rho d\phi d\theta$
- h. $\int_0^{\pi/2} \int_0^{\pi/4} \int_0^{18} \rho^4 \sin\phi d\rho d\phi d\theta$

17.

Let $y = 3u^u + 4u$ and $u = (7 - 3x)^6$. Evaluate $\frac{dy}{dx}$ when $x = 2$.

Answer:

18.

Use Newton-Raphson Method with initial point $(1, -1)$ and perform one iteration to find an approximation solution to the system:

$$\begin{cases} y^2 e^x = 3 \\ 2ye^x + 10y^4 = 0. \end{cases}$$

What is the x-coordinate of this approximation solution? (Round your answer to 2 decimal places.)

Answer:

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Answer ALL FOUR questions.

Notes:

- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- You must start each question on a new page. You should write down the question number on the top right hand corner of each page.
- If your answer to a question spans over more than one page, you must indicate clearly on each page (except the last) that the answer will continue on the subsequent page.
- Some formulae you may need:

For $n \geq 2$:

$$\begin{aligned}
\int \cos^n \theta \, d\theta &= \frac{1}{n}\cos^{n-1}\theta \sin\theta + \frac{n-1}{n}\int \cos^{n-2}\theta \, d\theta \\
\int \sin^n \theta \, d\theta &= -\frac{1}{n}\sin^{n-1}\theta \cos\theta + \frac{n-1}{n}\int \sin^{n-2}\theta \, d\theta \\
(\arctan x)' &= \frac{1}{1+x^2}
\end{aligned}$$

1. (10 marks) Find the absolute minimum and absolute maximum of $f(x, y) = 2x^2 - y^2 + 6y$ on the disc $x^2 + y^2 \leq 16$.

2. (10 marks) Find the area of the ellipse formed by the intersection of the plane $x + y - z = 1$ and the cylinder $x^2 + y^2 = 1$.

3. (10 marks) Consider the triple integral $\iiint_E x^2 dV$, where E is the region inside $x^2 + y^2 + z^2 = 36$ and $z = -\sqrt{3(x^2 + y^2)}$.

   (a) (3 marks) Transform the triple integral by using cylindrical coordinates.

   (b) (3 marks) Transform the triple integral by using spherical coordinates.

   (c) (4 marks) Calculate the result by either (a) or (b).

4. (16 marks) Let $\mathbf{x}$ be a vector in $\mathbb{R}^n$, A be an $n \times n$ symmetric matrix, and the function $f(\mathbf{x}) = \mathbf{x}^T A\mathbf{x}$ where $||\mathbf{x}|| = 1$.

   (a) (5 marks) Prove that if $\mathbf{x}$ is an eigenvector of A, $f(\mathbf{x})$ equals to the corresponding eigenvalue.

   (b) Let $\mathbf{x} = (x_1, x_2, x_3)^T$ and $f(\mathbf{x}) = \mathbf{x}^T A\mathbf{x} = 3x_1^2 + x_2^2 + 3x_3^2 + 4x_1x_2 + 4x_2x_3$.

      (i) (3 marks) Find the symmetric matrix A.

      (ii) (8 marks) If $x_1^2 + x_2^2 + x_3^2 = 1$, the maximum (minimum) eigenvalue of matrix A will be the maximum (minimum) value of $f(\mathbf{x})$. Find the maximum and minimum values of $f(\mathbf{x})$ subject to $x_1^2 + x_2^2 + x_3^2 = 1$, and find all the $\mathbf{x}$ when $f(\mathbf{x})$ achieves its maximum and minimum values.

******* **END OF PAPER** *******